

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# PERCENTAGES, (PERCENTAGE OF AN AMOUNT)

**LO: I can find a percentage of an amount using mental and written methods.**

## Finding a percentage of an amount

**Percentage** means 'out of 100'. To find a percentage of an amount, convert the percentage to a **decimal or fraction** and multiply.

**Mental shortcuts:** 10% = divide by 10; 5% = half of 10%; 1% = divide by 100. Build other percentages from these.

## Quick examples

|   |                  |                                 |
|---|------------------|---------------------------------|
| ● | 10% of £60 = £6  | divide by 10                    |
| ● | 25% of £80 = £20 | divide by 4                     |
| ● | 15% of £40 = £6  | 10% + 5% = 4 + 2 = 6            |
| ● | 10% of £60 = £60 | should divide by 10             |
| ● | 50% of 30 = 60   | 50% means half, so answer is 15 |

## Key idea

Percentage of amount = (percentage ÷ 100) × amount. Build complex percentages from 10%, 5%, and 1%.

$$\text{15\% of 80} = \text{10\% + 5\%} = 8 + 4 = 12$$

**Break percentages into easy parts**

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - using division by 10

Find 10% of £250.

10% means divide by 10

$$250 \div 10 = £25$$

**£25**

10% always means one tenth of the amount.

### Example 2 - building from 10%

Find 35% of £80.

$$10\% \text{ of } 80 = 8$$

$$30\% = 3 \times 8 = 24; 5\% = 8 \div 2 = 4$$

$$35\% = 24 + 4 = £28$$

**£28**

Build up from 10% and 5% for efficient mental methods.

### Example 3 - using a decimal

Find 17.5% of £240.

$$\text{Convert: } 17.5\% = 0.175$$

$$240 \times 0.175 = £42$$

**£42**

For awkward percentages, convert to a decimal and multiply.

### Example 4 - percentage as a fraction

Find 75% of 360.

$$75\% = \frac{3}{4}$$

$$360 \div 4 = 90$$

$$90 \times 3 = 270$$

**270**

75% = three-quarters. Divide by 4 then multiply by 3.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. 10% and multiples

Find each percentage mentally.

- a) 10% of £90
- b) 20% of £150
- c) 30% of £200
- d) 40% of £350

### 2. 5% and 1%

Use 10% to help find these.

- a) 5% of £60
- b) 15% of £80
- c) 1% of £400
- d) 11% of £300

### 3. Using decimals

Convert to a decimal and calculate.

- a) 12% of £250
- b) 8% of £75
- c) 17.5% of £160
- d) 2.5% of £800

### 4. Fraction method

Use the fraction equivalent to calculate.

- a) 25% of 180
- b) 50% of 94
- c) 75% of 240
- d)  $33\frac{1}{3}\%$  of 270

### 5. Real-world problems

Solve each problem.

- a) A class has 30 pupils. 60% are girls. How many girls?
- b) A £45 book has 20% off. What is the discount?
- c) 12.5% of 480 students got A\*. How many?
- d) A tip is 15% of a £65 meal. How much?
- e) VAT at 20% on £85. Find the VAT amount.

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) 10% of an amount means divide by 10 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) 25% of 60 = 20 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) 50% means half ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) 1% of 500 = 50 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) 15% = 10% + 5% ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A school has **1200 pupils**. 55% are boys. Of the boys, **40%** play football. How many boys play football?

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) A jacket costs **£85**. In a sale it is reduced by **15%**. A loyalty card gives an extra **5%** off the sale price. What is the final price?

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

